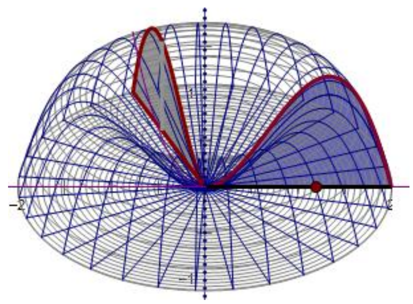
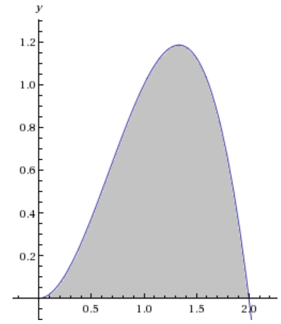
MTH 201 Chapter 1: Application of Integration

1.3 Volumes by Revolution: Shell Method

In 1.2 we found that we could compute the volumes of solids or revolution using the *washer* and *disk methods*. However, some volume problems are difficult to hand using these methods.

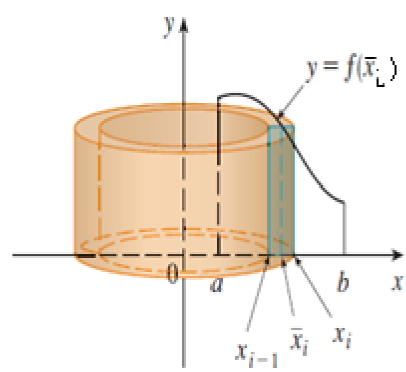
<u>Motivation:</u> Find the volume when the region enclosed by the curves $y = 2x^2 - x^3$ and $y = 0$ is rotated about the y-axis.

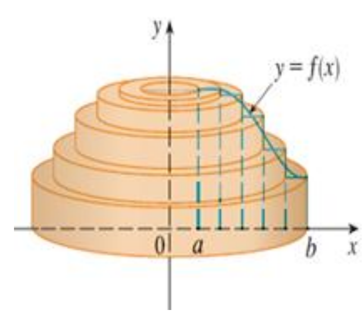




The following link helps to illustrate the *Cylindrical Shell Method*.
http://mathdemos.org/mathdemos/shellmethod/

Now, let's see if we can generate a Riemann sum using cylindrical shells instead of disks. Consider the picture below:





Cylindrical Shell Method

<u>Volume of a Solid of Revolution (w/ a *vertical* axis of rotation)</u>
If S is a solid that lies between $x = a$ & $x = b$ whose volume can be computed using cylindrical shells, then the volume of S is given by $V = \int_a^b 2\pi r h \, dx$
where r is the radius of any cylindrical shell and h is the height of any cylindrical shell. Both r and h are functions of x.

<u>Volume of a Solid of Revolution (w/ a *horizontal* axis of rotation)</u>
If S is a solid that lies between $y = a$ & $y = b$ whose volume can be computed using cylindrical shells, then the volume of S is given by $V = \int_a^b 2\pi r h \, dy$
where r is the radius of any cylindrical shell and h is the height of any cylindrical shell. Both r and h are functions of y.

<u>Example 1:</u> Find the volume when the region enclosed by the curves $y = 2x^2 - x^3$ and $y = 0$ is rotated about the y-axis.

Example 2: Suppose S is the solid obtained by revolving the region enclosed by $y = x$ and $y = x^2$ about the y-axis.

a) Find the exact volume of S by using cylindrical shells.

b) Find the exact volume of S by using washers/disks.

Note:

- The *<u>shell</u>* will always be *<u>parallel</u>* to the axis of rotation.
- Therefore if "washers/disks are *dy*", then "shells are *dx*" (and vice-versa).

<u>Example 3:</u> Use cylindrical shells to find the volume when the region enclosed by the curves $y=\sqrt{x}$, $y=0$, & $x=1$ is rotated about the *x*-axis.

<u>Example 4:</u> Find the volume when the region enclosed by the curves $y=x-x^2$ and $y=0$ is rotated about the line $x=2$.